

SOLVING CONTEXT SHUFFLE WITH OPENSEARCH

MITIGATING PERMUTATION-INDUCED HALLUCINATIONS IN PRODUCTION

Ramya Ravi

 [@ramya-ravi19](#)

✕ [@ramya_ravi19](#)

David vonThenen

    ✕ [@davidvonthenen](#)

INTRODUCTIONS

Ramya Ravi



- TODO TODO TODO
- TODO TODO TODO
- TODO TODO TODO
- TODO TODO TODO
- TODO TODO TODO
- TODO TODO TODO

 [@ramya-ravi19](#)

 [@ramya_ravi19](#)

David vonThenen



- Agentic Solutions
- Book In Progress
- AI Harness Wizard
- NLP Sorcery
- AI/ML Security Hacker
- Part-time GPU Owner

     [@davidvonthenen](#)

AGENDA

- **Research Paper: Use Prompt Repetition?**
- **Deconstructing RAG Prompts**
- **Discuss: "Adversarial Prompting"**
- **Prompt Evaluation**
- **Q & A**

RESEARCH PAPER DISCUSSION

PROMPT REPETITION IMPROVES NON-REASONING LLMs

THE RESEARCH PAPER...

Link:
<https://arxiv.org/abs/2512.14982>



 Search [Submit](#) [Donate](#) | [Log in](#)

Computer Science > Machine Learning

[Submitted on 17 Dec 2025]

Prompt Repetition Improves Non-Reasoning LLMs

Yaniv Leviathan, Matan Kalman, Yossi Matias

When not using reasoning, repeating the input prompt improves performance for popular models (Gemini, GPT, Claude, and Deepseek) without increasing the number of generated tokens or latency.

Subjects: **Machine Learning (cs.LG)**; Artificial Intelligence (cs.AI); Computation and Language (cs.CL)

Cite as: [arXiv:2512.14982](https://arxiv.org/abs/2512.14982) [cs.LG]

(or [arXiv:2512.14982v1](https://arxiv.org/abs/2512.14982v1) [cs.LG] for this version)

<https://doi.org/10.48550/arXiv.2512.14982> 

Submission history

From: Matan Kalman [[view email](#)]

[v1] Wed, 17 Dec 2025 00:33:51 UTC (1,403 KB)

Access Paper:

[View PDF](#)

[HTML \(experimental\)](#)

[TeX Source](#)



[view license](#)

Current browse context:
cs.LG

[< prev](#) | [next >](#)
[new](#) | [recent](#) | [2025-12](#)

Change to browse by:

[cs](#)
[cs.AI](#)
[cs.CL](#)

References & Citations

[NASA ADS](#)
[Google Scholar](#)
[Semantic Scholar](#)

DEMO: PROMPT REPETITION

IN MULTIPLE WAYS...

RESULTS

● Qwen 3.5 9B

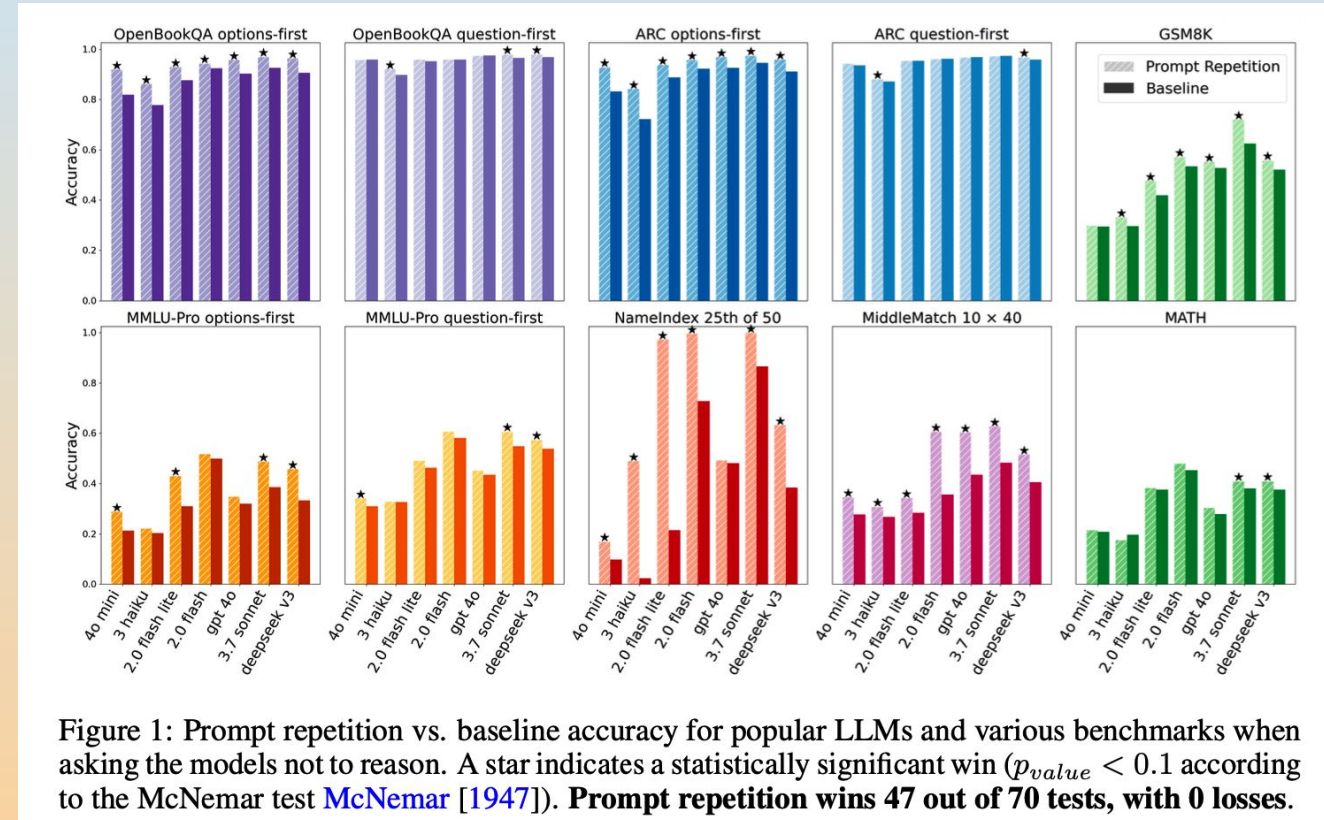
- Disable Reasoning: Wrong
- Enable Reasoning: Correct

● OpenAI

- GPT-4o: Not Available 😊
- GPT-4o-mini: NA 😊

● Claude

- Sonnet 3.7: Not Available 😊
- Haiku 3: NA 😊



Links:

Qwen 3.5 9B GGUF - <https://huggingface.co/mlx-community/Qwen3.5-9B-MLX-4bit/tree/main>

Qwen 3.5 9B MLX (Apple) - <https://huggingface.co/unsloth/Qwen3.5-9B-GGUF/tree/main>

WHY DOES THIS WORK?

- **Language Models (LMs) Are Language Based**

- An (Non-Reasoning) LLM Reads From Left to Right
- Context Follow By Question, What is Relevant? Fail!
 - Prompt Repetition Works, Gives a 2nd Pass
- This is Why Reasoning LLMs Are Unaffected
 - At Cost of (Reasoning) Tokens & Your Wallet!

- **How Does This Affect RAG?**

- Permutation-Induced Hallucinations
- Long Context and Prefill Efficiency

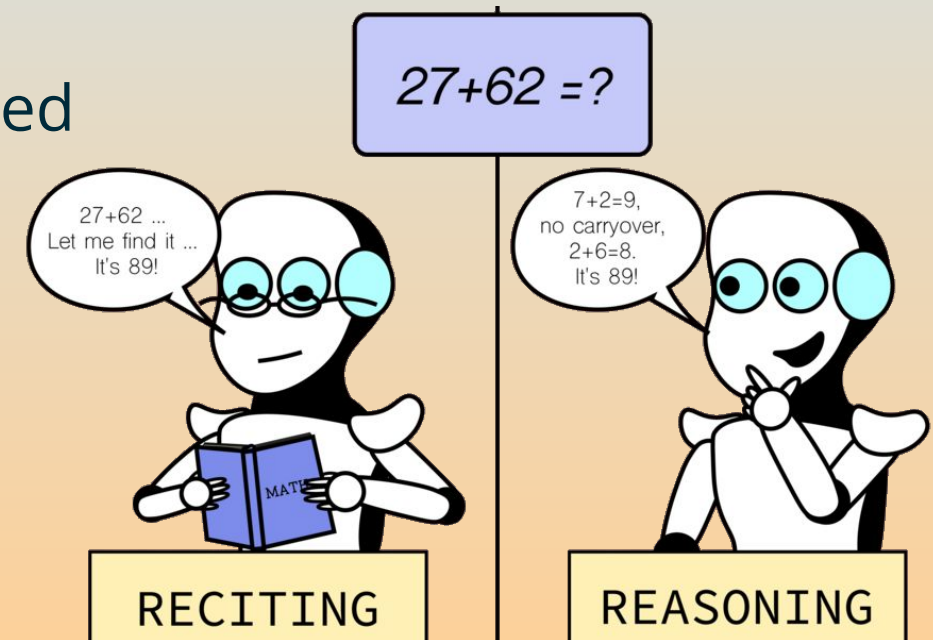


Image Attribution:

Reasoning skills of large language models are often overestimated

<https://news.mit.edu/2024/reasoning-skills-large-language-models-often-overestimated-0711>

RECONSTRUCTING RAG PROMPTS

RESEARCH-BASED PROMPT STRUCTURE FOR RAG

TODO

- **TODO**
 - TODO
 - TODO

"ADVERSARIAL PROMPTS"

**WHAT PROMPT INJECTION TEACHES US ABOUT BAD
STRUCTURE**

WHAT ARE ADVERSARIAL ATTACKS?

- **Intentional Manipulation to Affect LM Outputs**
 - In Security, It's Called Pentesting (Penetration Testing)
 - "How do I Hack This to do XYZ?"
 - With LM, Language is VERY Ambiguous!
- **Basic Examples:**
 - Manipulating Classifiers
 - In Language, [Sentiment Analysis](#)
 - In Visual, [Block Person Identification](#)
- **[WeAreDevelopers Video: Confuse, Obfuscate, Disrupt](#)**
 - Apply Pentesting to Get Results/Output!



LET'S START WITH COMMON MISTAKES ...

• Cross-Chunk Bleeding

- Bounded Data or Info Without Delimiters

<background></background>

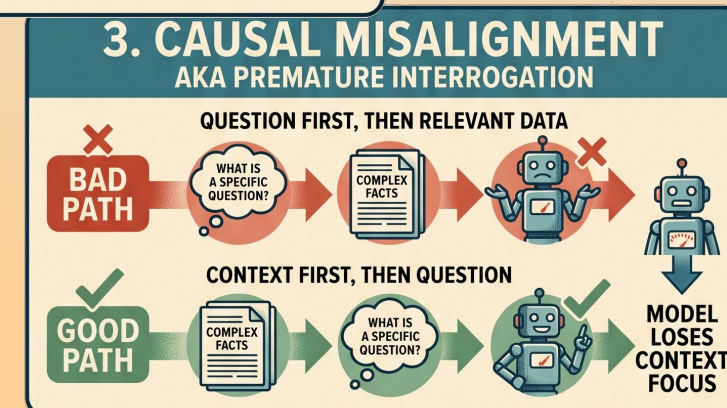
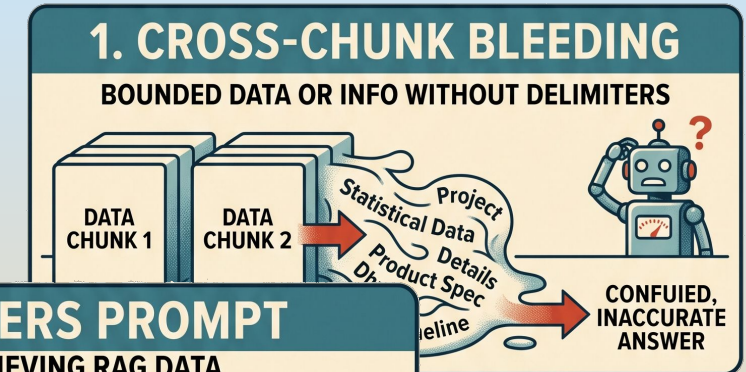
<datachunk2></datachunk2>

• Hoarders Prompt

- Over Retrieving RAG Data
- Top-K of 10 Might Be Too Much
- Use Reranker for Relevance Drop-off

• Causal Misalignment

- Aka Premature Interrogation
- BAD: Question First, Then Relevant Data
- Think as an Auto-Regressive LLM



EXAMPLE: JSON OR RAW PAYLOAD DUMP

- Remember, LLMs Are **LANGUAGE** Based!

- AI Engineers Get "Lazy" Attach Raw Data
- Using JSON Means Substantial Token Bloat
 - Interfers with Chain-of-Thought

- Research: Inc. 5-20% Failure Rates

- How To Fix This? (Platform Specific)

- Heavy Preference for XML
- Alternatives: CSV (not joking), TOON
- Upshot: Use Human-Readable Formats!

License: CC BY-SA 4.0
arXiv:2408.02442v1 [cs.CL] 05 Aug 2024

Let Me Speak Freely? A Study on the Impact of Format Restrictions on Performance of Large Language Models

Zhi Rui Tam
Equal contribution Appier AI Research

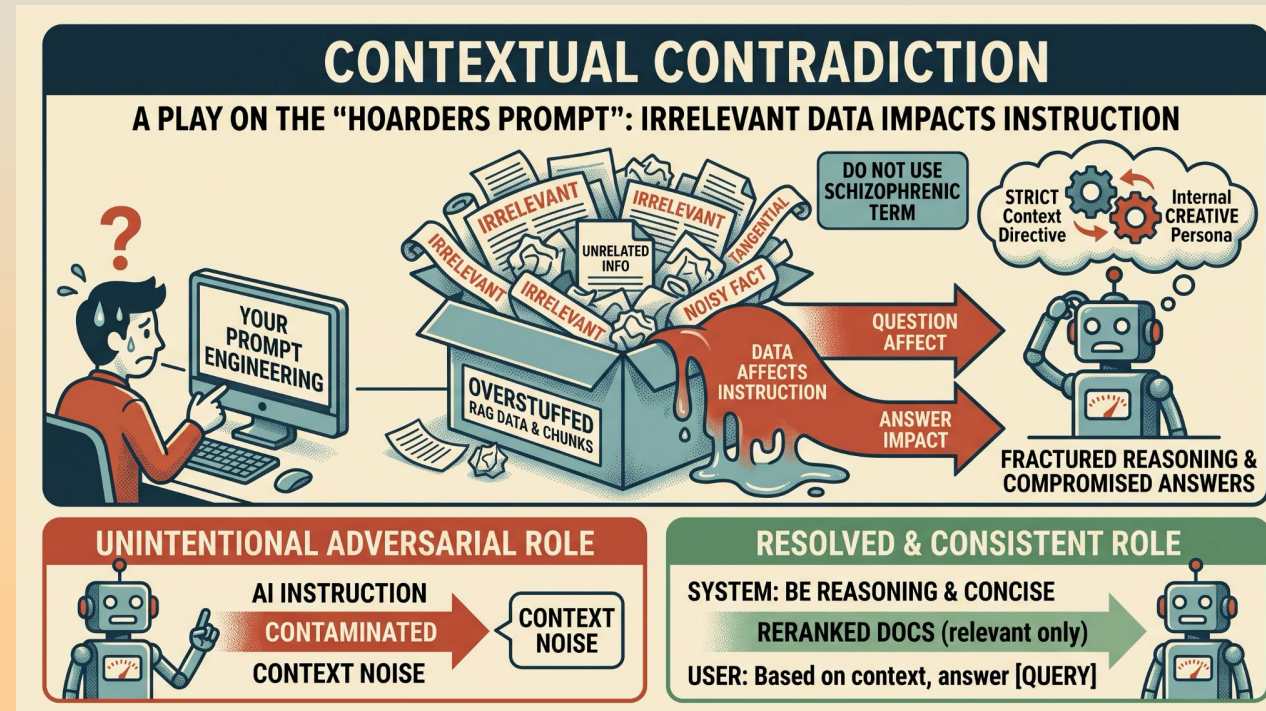
Cheng-Kuang Wu ¹
Appier AI Research

Chieh-Yen Lin
Appier AI Research

Hung-yi Lee
Equal advisorship National Taiwan University

EXAMPLE: CONTEXTUAL CONTRADICTION

- Aka the "Schizophrenic" Prompt or Data
- How This Happens?
 - RAG Chunks Are Irrelevant to Question
 - Data Effects Question & Answer
 - Even Irrelevant Data
 - Play On Hoarders Prompt
 - Too Much Info



DEMO: CONTEXTUAL CONTRADICTION

THE BUFFET OF AMBIGUITY

REVIEW: How We Got Here

- **Not Formatting Our Data for LMs**

- Cross-Chunk Bleeding
- JSON or Raw Payload Dump

- **Poor Data Quality**

- Hoarders Prompt
- Contextual Contradiction

- **Digestion or Ordering**

- Causal Misalignment



This is TRUE Garbage In, Garbage Out

Image Attribution:

GIGO — Garbage in Garbage Out Concept for UX Research

<https://medium.com/nyc-design/gigo-garbage-in-garbage-out-concept-for-ux-research-7e3f50695b82>

PROMPT EVALUATION

How Do You EVALUATE WHAT "GOOD" Is?

TODO

- TODO
 - TODO
 - TODO

SESSION RESOURCES

RESOURCES

SCAN THE QR CODE FOR ALL RESOURCES!!

github.com/instaclustr/2026-opensearchcon (TODO; URL?)

TODO

TODO

Check Out NetApp Instaclustr:

- OpenSearch: Vector and BM25 for RAG
- Kafka: Data Streaming and Event-Driven Arch
- Cassandra: Hybrid RAG
- ClickHouse: AI Governance
 - Observability, Auditing, Metrics



THANK YOU

Ramya Ravi

 [@ramya-ravi19](#)

✕ [@ramya_ravi19](#)

David vonThenen

    ✕ [@davidvonthenen](#)